INDUJA M K

indujamk2005@gmail.com — +91 94435 51896 — linkedin.com/in/induja-m-k-b9849631a

PROFESSIONAL SUMMARY

Electrical and Electronics Engineering undergraduate with strong fundamentals in Digital Electronics, Analog Electronics, Embedded Systems, and SoC communication protocols. Hands-on experience in Embedded C/C++, Arduino, ESP32, firmware development, hardware validation, and IoT-based system design. Familiar with UART, SPI, I2C, ADC/DAC interfacing, CMOS fundamentals, Op-Amps, and PCB design. Strong analytical, problem-solving, teamwork, and communication skills with a keen interest in Semiconductor Technologies.

EDUCATION

B.E. Electrical and Electronics Engineering — **CGPA: 8.19**
K.S.R. College of Engineering — 2023–2027

Higher Secondary Certificate — **80.33%**
S.V.V Hr. Sec School

Secondary School Leaving Certificate — **80.22%**
Sarawathi Vidhyashram Matric Hr. Sec School

TECHNICAL SKILLS

Programming	C, Embedded C, C++
Digital Design	Logic Gates, Combinational & Sequential Circuits, Flip-Flops
Embedded Systems	Arduino, ESP32, 8051 Microcontroller
Communication Protocols	UART, SPI, I2C, Wi-Fi, Bluetooth
Analog & Mixed Signal	Op-Amp, ADC/DAC, CMOS Basics, Analog Electronics
Tools	Keil IDE, Arduino IDE, Proteus, Eagle PCB, Wokwi, Tinkercad
Core Areas	Embedded Systems, SoC Fundamentals, Microprocessor Fundamentals, PCB Design, IoT, VLSI Fundamentals
Soft Skills	Problem Solving, Team Collaboration, Verbal and Written Communication

INTERNSHIP EXPERIENCE

Embedded Systems Intern — Msquare Automation

- Developed and tested microcontroller firmware using Embedded C for GPIO, ADC, and PWM-based applications.
- Applied structured testing methodologies to improve hardware reliability and integration performance.

Embedded Development Intern — Apex I SYS

- Executed structured testing and troubleshooting of embedded applications to improve system reliability.
- Validated peripheral communication and hardware-software integration in real-time embedded systems.

PROJECTS

Automatic Smart Parking System

- Designed an IR sensor-based vehicle detection system with automated servo-controlled gate operation using Arduino.
- Implemented real-time sensing, control logic, and hardware validation for reliable operation.

IoT-Based Automatic Headlight Dimmer

- Developed an ESP32-based intelligent headlight control system using sensor inputs and automatic intensity adjustment.
- Integrated embedded firmware, hardware interfacing, and wireless monitoring concepts.

CERTIFICATIONS

- NPTEL – Introduction to Industry 4.0 and Industrial Internet of Things (Elite)
- Embedded Systems Training – Internshala (8 Weeks)
- Value Added Course – Embedded SoC Protocol
- PCB Design and Fabrication
- AICTE AI/ML Virtual Internship
- Workshop on Keil Software

LEADERSHIP & ACTIVITIES

- Master of Ceremonies for Technical Symposium and Department Events.
- Demonstrated teamwork, communication, leadership, and event coordination skills.